A SCHEME WHOSE UNIT MAP TO THE SPECTRUM OF ITS GLOBAL SECTIONS SPLITS IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 23546-23552 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 23554-23563, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Tiny categorical affineness criterion; used in the chapter's terminal affineness arguments.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Schemes, lemma morphism-into-affine (the canonical map and its universal property); the anti-equivalence of affines and rings (fully faithfulness of Spec).

Source reference. Original lemma 01I1: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 23563–23563. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let X be a scheme. If the canonical morphism $X \to \mathrm{Spec}(\Gamma(X, \mathcal{O}_X))$ of Schemes, Lemma 01I1 has a retraction, then X is an affine scheme.*

Proof. Write $S = \mathrm{Spec}(\Gamma(X, \mathcal{O}_X))$ and $f : X \to S$ the morphism given in the lemma. Let $s : S \to X$ be a retraction; so $\mathrm{id}_X = sf$. Then $fsf = \mathrm{id}_S f$. Since f induces an isomorphism $\Gamma(S, \mathcal{O}_S) \to \Gamma(X, \mathcal{O}_X)$ this means that fs and id_S induce the same map on $\Gamma(S, \mathcal{O}_S)$. Whence $fs = \mathrm{id}_S$ as S is affine. Hence f is an isomorphism and X is an affine scheme, as was to be shown. □

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.